Learning To See In The Dark - Enhancement

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Abstract

Low-light image enhancement under extreme illumination is difficult because short-exposure RAW images often suffer from severe noise, low visibility, and missing details. This project studies a RAW image restoration framework using paired short- and long-exposure Sony RAW images, aiming to map severely underexposed inputs to high-quality RGB outputs. Based on conditional U-Net, we propose a Bridge U-Net model, and further explore two fine-tuning settings and few-shot meta-learning from a pretrained checkpoint, with DDBM as a comparison model. Quantitative metrics and visual results show the potential of bridge-based feature refinement for extreme low-light RAW image enhancement.

1 Introduction

Low-light image restoration is an important problem in computer vision, with applications in night photography, surveillance, drones, and search-and-rescue. Under extreme illumination, short-exposure RAW images often suffer from severe noise, underexposure, low visibility, color distortion, and missing scene details. Although long-exposure images can improve brightness, they may introduce blur and are often impractical in real-world scenes. Therefore, restoring high-quality images from short-exposure RAW inputs remains a challenging task.

In this project, we study extreme low-light RAW image restoration using a Bridge U-Net based framework. We introduce bridge refinement, FiLM, and teacher latent supervision, and further explore fine-tuning, few-shot meta-learning, and DDBM for comparison. The goal is to evaluate whether bridge-based latent refinement can improve restoration quality under limited-resource conditions.

2 Related Work

Low-light image restoration under extreme conditions has been studied in prior work. The most relevant study is *Learning to See in the Dark* by Chen et al. (2018), which proposed an end-to-end framework for mapping short-exposure RAW images to high-quality long-exposure outputs and established a strong U-Net based baseline. However, it does not explicitly explore latent feature refinement or adaptation under limited-data settings. These limitations motivate our bridge-based design, as well as the exploration of fine-tuning, few-shot learning, and DDBM for comparison.

3 Methodology

This project studies extreme low-light image restoration using paired short- and long-exposure Sony RAW images. Our framework is based on U-Net, with bridge refinement, FiLM, and teacher latent supervision. Based on this framework, we further explore full supervised training, two fine-tuning settings, few-shot meta-learning, and DDBM comparison.

3.1 Data Preprocessing

The data pipeline uses paired Sony RAW images, each consisting of a short-exposure input and a corresponding long-exposure reference. The RAW data are converted into a four-channel packed Bayer representation after black-level subtraction and normalization. The short-exposure input is scaled by the exposure ratio, clipped at 300, to better match the target dynamic range, while the long-exposure RAW image is processed into an RGB supervision target with consistent camera settings.

3.2 Main Model

Since in this project, the input images have some problems: severe noise, color distortion, and substantial loss of scene information. This makes the reconstruction task considerably more challenging than standard image enhancement [1]. To address these challenges, we extend the original U-Net by adding a bridge module between the encoder and decoder. This bridge refinement stage improves degraded latent features before reconstruction, helping the decoder recover cleaner structures, more accurate colors, and missing scene details from low-light RAW inputs.

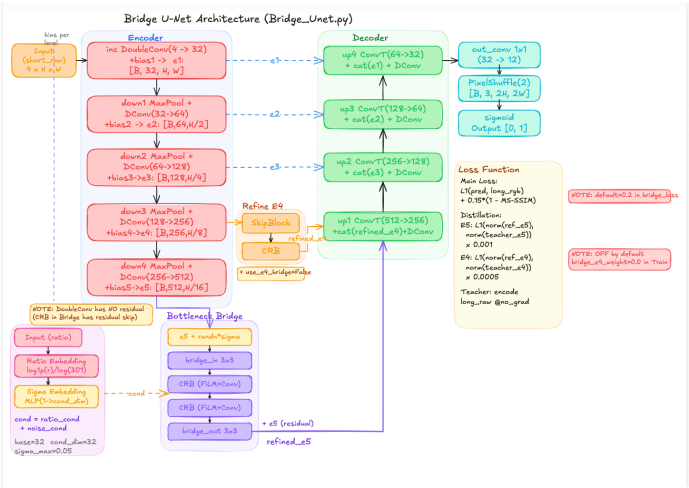


Figure 1: Main Bridge Model

In addition, we retain the skip connections from the standard U-Net to combine encoder and decoder features, allowing the decoder to recover fine spatial details that may be lost during down-sampling [2]. Unlike the traditional skip connection, we apply the conditional refinement step to the e4 skip features, which makes the model to adaptively suppress noise in skip features before they reach the decoder.

We introduced a method called FiLM to enhance our Bridge UNet model. FiLM stands for Feature-wise Linear Modulation, which applies a feature-wise affine transformation to intermediate features based on conditioning information [3]. In our implementation, the condition vector is obtained from the exposure ratio and noise level, and is used to modulate latent features in the bridge module through channel-wise scaling and shifting. This allows the model to adjust its feature refinement process adaptively for different degradation levels in extreme low-light RAW images.

Furthermore, we introduced teacher latent distillation to further regularize the refinement process in the bridge module. This design is motivated by knowledge distillation, where the knowledge learned by a teacher model is transferred to a student model during training [4]. In our model, the long-exposure branch provides teacher features, while the bridge-refined features act as student features, following feature-based distillation similar to FitNets [5]. During training, the long-exposure RAW image is passed through the shared encoder to extract teacher features at e4 and e5, while the bridge-refined features are used as student features. After L2 normalization, an L1 loss is computed between them, encouraging the refined latent features to align with those from well-exposed, less noisy inputs.

3.3 Fine Tuning

For the fine tuning stage, we adapt the Bridge U-Net model previously trained on SID to the ELD dataset. The code loads the Stage-1 checkpoint and initializes the same `BridgeUnet` architecture with four RAW Bayer input channels and three RGB output channels. The ELD dataset is built from short-exposure and long-exposure RAW pairs captured with different cameras, including Nikon D850, Canon EOS 70D, and Canon EOS 700D. Each short RAW image is packed into a four-channel Bayer representation, while the corresponding long-exposure RAW image is demosaiced into an RGB ground truth. The code also follows the ELD exposure protocol by assigning ratios of x1, x10, x100, and x200 according to the image index inside each scene. During training, aligned random crops are extracted using 512×512 patches for the RAW input and 1024×1024 patches for the RGB target, with horizontal/vertical flips and transpositions applied for augmentation. For validation, one scene per camera is held out and evaluated using centered crops without random augmentation.

The fine-tuning process updates the entire model using AdamW, without applying selective freezing, meaning that all trainable layers of the Bridge U-Net are optimized together for the new camera domains. The main reconstruction objective is an L1 loss between the predicted RGB image and the long-exposure RGB target. After a fixed starting epoch, an MS-SSIM term is progressively added to improve structural similarity and reconstruction quality. The training script also includes mixed precision, gradient clipping, cosine learning-rate decay, outlier-loss skipping, and an exponential moving average of the model weights to make training and validation more stable. After each epoch, the model is evaluated using PSNR, SSIM, and MS-SSIM, and the best checkpoint is saved when the validation PSNR improves. Overall, this fine-tuning stage transfers the SID-trained model to Nikon and Canon RAW images while keeping the same Bridge U-Net architecture and training it end-to-end.

3.4 Few Shot

This project wants to know whether can still manage the task if the training data is limited. So, we explore few-shot learning by using the meta-learning, which is an important strategy to solve the few-shot learning. Few shot means that the model generalize by using limited training examples. Meta-learning, also known as learning to learn, focuses on acquiring transferable priors from previous tasks so that the model can adapt more efficiently to new tasks [6].

We used the pretrained Bridge Unet as the base for building our meta-learning. In our model, each scene is treated as a task. In the 5-shot setting, five samples are used as the support set, and up to 8 additional samples from the same scene are used as the query set. For each task, the model performs 3 inner-loop gradient updates on the support set, then evaluates the adapted model on the query set, whose loss is used for outer-loop optimization. The inner-loop adaptation is implemented with the higher library, and consistent random augmentation is applied during training.

3.5 Denoising Diffusion Bridge Models (DDBMs) - Inspired Feature Bridge

To explore whether the diffusion-based bridge method can further improve the restoration quality, We are inspired by the ideas of DDBM and build a simplified version to test the ideas. DDBM defines a Brownian bridge process between paired endpoints and learns a score function to reverse the diffusion process for sampling. In the adaptation, we keep the Brownian bridge interpolation formulation but use it in the feature space of our U-Net encoder rather than in pixel space. We then replace the score-matching objective with a direct reconstruction loss.

The reason why we made this change, is using a full pixel-space resolution will require a large amount of compute and our inference sampler is a *deterministic self-refinement* that reuses the encoder as a pseudo-long feature extractor, which keeps the sampler single-network and memory-light at the cost of sampling flexibility.

Architecture: The DDBM-BridgeUnet shares the same encoder structure as the main Bridge U-Net (Section 3.2), but introduces two key modifications to the decoder:

Time-conditioned decoding We take the bridge parameter t in, and turn it into a sinusoidal vector just like the diffusion models do for time steps, and then inject the vector into every decoder block. This informs the decoder how far the current features are between the short-exposure and the long-exposure.

Selective bridge interpolation

We only apply the bridge at the 3 deepest levels c_3, c_4, c_5, and keep the shallow features c_1, c_2 from the short-exposure branch to preserve spatial structure. Given short features f_{short} and long features f_{long}, the interpolated feature at time $t \in [0, 1]$ is:

$$f_t = (1-t) \cdot f_{\text{short}} + t \cdot f_{\text{long}} + \sigma_{\max}\sqrt{t(1-t)}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{1}$$

with $\sigma_{\max} = 0.1$ and $t \sim \mathcal{U}(0.01, 0.99)$.

Training objective The network predicts the clean target (i.e. x_1-prediction):

$$\mathcal{L} = \mathcal{L}_{L1}(\hat{y}, y) + 0.1\big(1 - \text{MS-SSIM}(\hat{y}, y)\big)$$

Iterative inference. Starting from the short-exposure features at $t = 0$, the decoder produces an initial estimate y_0. At step $k = 1, \ldots, N$, y_{k-1} is down-sampled and re-encoded as pseudo-long features, deterministically interpolated with the fixed short-exposure features at $t_k = k/N$, and decoded with the corresponding time embedding. We use $N = 3$ steps at evaluation.

Advantages The x_1-prediction objective preserves image structure(basic settings) and low-level details. Training becomes stable under supervised learning with a single forward pass at $t = 0$ produces a usable output, computation cost is cheaper than usual reverse-SDE Sampler

Disadvantages Since our model does not learn score function, so it cannot simulate the reverse bridge dynamics stochastically; it is a deterministic feature-space approximation.

4 Analysis

In this experiment, we use three metrics to analyze the model: L1 loss, PSNR (Peak Signal-to-Noise Ratio), and MS-SSIM.

L1 loss is used to measure the mean absolute error (MAE) between the predicted image $\hat{I}$ and the ground-truth image I:

$$\mathcal{L}_{L1} = \frac{1}{N}\sum_{i=1}^{N} |\hat{I}_i - I_i|,$$

where N denotes the total number of elements in the output image. This metric reflects the average pixel-wise reconstruction error, and lower values indicate that the predicted result is closer to the reference image at the pixel level.

PSNR is a standard full-reference metric that measures reconstruction fidelity based on the mean squared error. Although it is reported in our experiments as an auxiliary indicator, it is not the primary metric of interest in this work, since PSNR does not fully capture perceptual quality such as structural fidelity and visual consistency. In particular, an image with a higher PSNR value does not necessarily appear more visually natural or structurally faithful.

MS-SSIM, or multi-scale structural similarity, is motivated by the observation that the perceptibility of image details depends on image resolution and viewing conditions. For two image patches x and y, MS-SSIM is defined as

$$\text{MS-SSIM}(x, y) = [l_M(x, y)]^{\alpha_M} \prod_{j=1}^{M} [c_j(x, y)]^{\beta_j} [s_j(x, y)]^{\gamma_j},$$

where M denotes the number of scales, $c_j(x, y)$ and $s_j(x, y)$ are the contrast and structure comparisons at scale j, and $l_M(x, y)$ is the luminance comparison computed only at the coarsest scale. MS-SSIM evaluates image similarity across multiple resolutions instead of measuring structural similarity at only a single scale [7]. Compared with PSNR, MS-SSIM better reflects human perception and structural fidelity, making it more suitable for evaluating this project.

4.1 Main model

This is the result of our main bridge model.

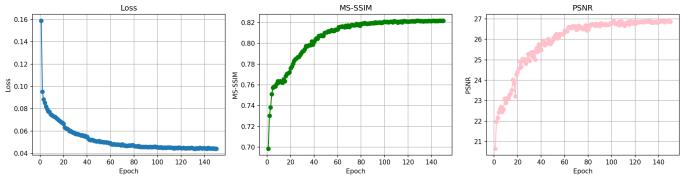


Figure 2: Main Model

From the above plot, the loss decreases steadily while MS-SSIM and PSNR increase and reach a stable plateau, which shows a convergent and stable training process. In the Loss plot, the value starting at 0.16 and decreases to 0.08 really quickly, then it then stabilized at around 0.044. MS-SSIM improves smoothly to above 0.82, showing good structural restoration, and PSNR increases to around 26.9 dB. Overall, the main Bridge model achieves reasonably good performance.

4.2 Fine Tuning

This is the result for fine tuning on decoder and bridge part.

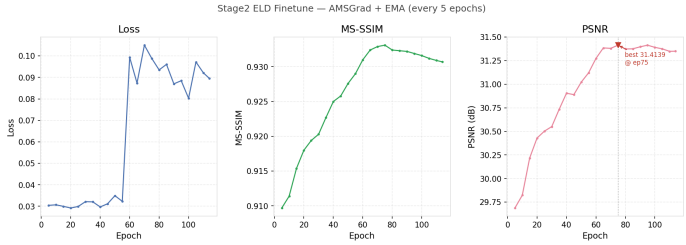


Figure 3: Fine Tuning

From the above result, we can see that the L1 loss curve shows a upward around epoch 60, which is unusual with the common loss plot. However, MS-SSIM and PSNR is increasing after that point and the value of MS-SSIM is higher than 0.90. The high value of MS-SSIM represents that the model has learned to preserve structural information well and can produce perceptually high-quality images. This suggests that the model performance is still improving, and the loss increase is likely due to changes in training behavior rather than real degradation.

4.3 Few Shot

This is the result of our meta learning result.

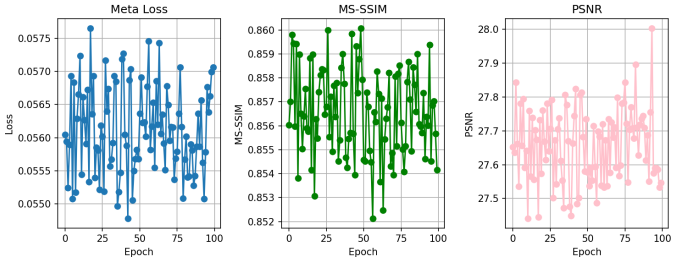


Figure 4: Meta Learning Model

The results show that L1 Loss, MS-SSIM, and PSNR remain within relatively narrow ranges with noticeable fluctuations. The meta-learning model stays stable during training without divergence, and MS-SSIM remains at a reasonably good level, indicating that the model still has effective

reconstruction capability. However, the improvement trend is not obvious, since the loss decreases only slightly and both MS-SSIM and PSNR mainly fluctuate within a small range. This suggests that the model has reached a stable plateau, likely because meta-learning is initialized from a strong pretrained checkpoint, leaving limited room for further improvement. Thus, the consistently good MS-SSIM and PSNR values still demonstrate the effectiveness of the meta-learning framework.

4.4 DDBM

This is the result of our DDBM-feature model and we have

Table 1: Comparison between canonical DDBM [8] and our feature-space variant.

	True DDBM	**Ours(DDBM - Feature)**
Objective	Learn score $\nabla \log p(x_t \mid x_1)$	Predict clean RGB directly
Loss	Bridge score matching	$L_1 + 0.1\,(1-\text{MS-SSIM})$
Inference	Reverse SDE ($25-1000$ steps)	Deterministic bridge ($N=3$)
Bridge	Pixel / latent state	Encoder features c_3-c_5

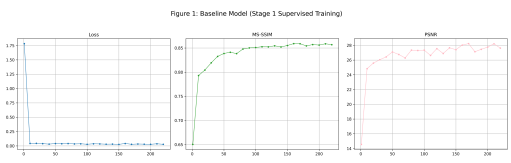


Figure 5: Baseline of DDBM

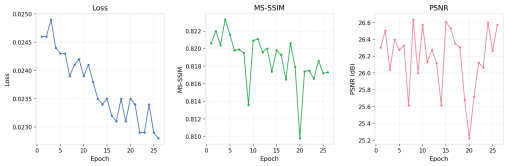


Figure 6: Stage 2 of DDBM

From the above result, we can see that the L1 loss of the baseline DDBM-feature model decreases rapidly and reaches a stable plateau in the early stage of training, while MS-SSIM and PSNR increase quickly and then also stabilize. Specifically, the L1 loss drops from around 1.75 to approximately 0.02, and MS-SSIM increases from about 0.65 to above 0.85, indicating that the model has learned effective feature representations. However, during Stage 2, the values of L1 loss, MS-SSIM, and PSNR fluctuate within a narrow range without an obvious upward trend. This suggests that the model enters a plateau early, and although it maintains a reasonable performance level, the overall improvement of DDBM is limited under the current setting.

On the SID Sony test split (598 photos), our DDBM-Feature model achieves **PSNR 26.12 dB** and **MS-SSIM 0.807** under $N = 3$ step iterative inference.

5 Result

This is the result of our models.



Figure 7: Final Result of Main Model

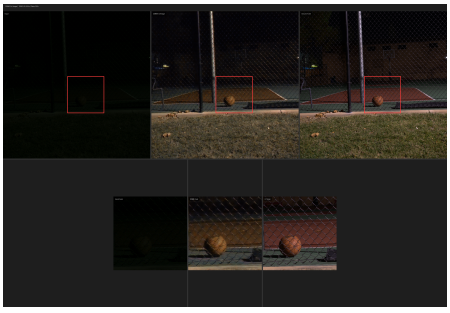


Figure 8: Final Result of DDBM

The output is much clearer than the extremely dark input, showing that the model has learned effective low-light enhancement. The few-shot model remains visually close to the main model, mainly because they are built from the pretrained Bridge model and their quantitative differences are very small. However, some details, such as the iron fence, remain unclear, indicating room for improvement. In this example, DDBM appears visually clearer than the main model, especially in the highlighted region. Thus, a reliable comparison should consider both visual results and quantitative metrics.

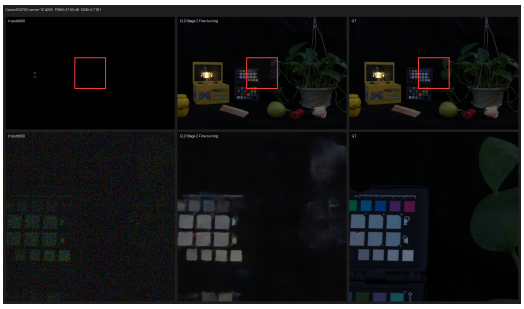


Figure 9: Scene 10, $\times 200$

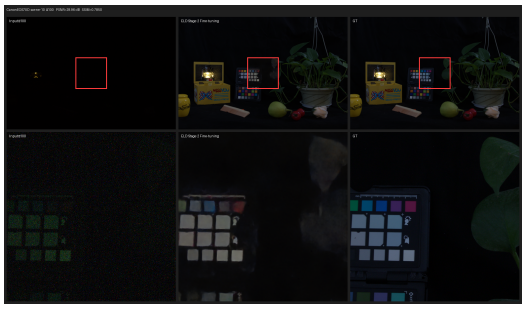


Figure 10: Scene 10, $\times 100$

We fine-tune our stage-1 DDBM-feature backbone on the ELD dataset proposed by Wei[9]. Initially we attempted to adapt our model on Fujifilm's X-trans architecture, but the result is not good due to the difference of color filter array from the cmos sensors. The model shows better adaptability on same Bayer Architecture. We select 2 photos from CanonEOS70D. Exposure ratio is defined as the ratio of the long exposure time to the short exposure time. In the this case, x100 brighter than x200. As a result, the model still can produces output, the the quality decreased as the exposure ratio increased.

6 Conclusion

In this project, we study low-light RAW image restoration under extreme conditions by using Bridge Unet, combined with fine-tuning, meta-learning, and DDBM. The Bridge Unet shows stable training and strong quantitative performance, while DDBM achieves comparable visual results. However, both are limited in fine detail recovery. In contrast, the fine-tuning model produces clearer results and better detail reconstruction, although its L1 loss curve shows an unusual trend. Overall, each model has its own strengths and limitations, and a comprehensive evaluation should consider both quantitative metrics and visual quality.

However, this project still has some limitations. The performance differences among the models are relatively small, leading to visually similar results. In addition, the fine-tuning model shows an unusual L1 loss trend that does not converge as expected, indicating the need for further investigation. In the future, we will explore more suitable models and evaluation strategies, and design more effective loss functions to better match perceptual quality.

7 Individual Contribution

Ruiqing Huo: Wrote the code and report about the main Bridge Unet model and few shot parts.
Payton Pei: Wrote report DDBM-Feature
Maxime Monast-Doré : Did Literature review and fine tuning part.

8 Usage AI

We used prompt "fix the text if necessary" and "fix the latex syntax error". We used artificial intelligence to identify the most relevant scientific articles and papers for our work.

References

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